

Data Mining, Big Data, and Analytics Project Document

Project Description:

You are required to find an innovative idea and a dataset that supports your idea, and on which you will apply the analytical techniques we use throughout the course.

Your idea must address a business problem, bring a business solution, or provide values, insights, or recommendations for business users; those who will benefit from the results of your work.

You are required to use one of the big data processing frameworks (e.g., Hadoop, Spark, Flink, etc.) integrated with your preferred language. The system should be implemented in pseudo-distributed mode (single-machine multiple processes). The implementation in fully distributed mode (multiple-machines multiple-processes) will be a bonus.

FAQ:

1. What do you mean by an innovative idea?

An innovative idea is a new business problem you are trying to solve. For example:

- A business problem for a bank may be “Should we give this customer a loan or not?” or “Based on what factors/conditions should we give our customers a loan?”
- A business question for a retail store concerning shelf management may be “What are the items commonly bought together by a sufficiently large number of customers?” This is commonly referred to as market basket analysis.
- A business problem for a magazine/newspaper may be “What determines a customer’s decision to subscribe or not?”

Your idea is not limited to business only but can be extended to governmental bodies, environmental institutions, and societal organizations. For example:

- “Which people are more likely to vote for/against a law?” 🗳️ “How will the climate on Earth change over the next ten years?”

2. Is there a restriction on the dataset?

Not really. You should find a dataset that’s large enough. We are talking here about datasets in order of megabytes and gigabytes.

3. What programming language(s) can I use?

This is an analytics project, so it is recommended to use either R or Python for this purpose. Both languages are very popular for data analytics. However, there is no real restriction on which programming languages to use.

4. What analytical techniques can I use?

You can use (but are not limited to) all the analytical techniques studied in this course.

Example:

- For a classification problem, you can build a classifier using logistic regression or K-NN, train a neural network, or build an SVM, among many other techniques.
- For a prediction problem, you can go with MLR (Multilinear Regression) or PCR (Principal Component Regression).
- For a segmentation problem, you can try different clustering techniques (K-means, hierarchical clustering, etc.)

5. We are seniors and about to graduate. Will the project consume too much time?

No.

6. I have no idea in my mind.

We recommend having a look at Kaggle and Analytics Vidhya, where you will find lots of ideas and datasets. You will also find many active competitions in which you can actually contribute and participate.

Deliverables:

1. Project Proposal:

The proposal should not exceed one page, specifying the following points clearly:

a. Idea:

The problem statement should be described clearly. **b.**

Dataset(s):

Links to the dataset(s) that will be used.

c. Planned approach or proposed solution.

A very brief plan of your proposed approach (you can change it later during the implementation phase).

2. Final Delivery:

a. Final Document containing:

- I. Brief problem description.
- II. Project pipeline.
- III. Analysis and solution of the problem:
 - 1) Data preprocessing.
 - 2) visualization.
 - 3) Extracting insights from data.
 - 4) Model/Classifier training.
- IV. Results and Evaluation.
 - 1) Model accuracy on train, test, and validation data.
- V. Unsuccessful trials that were not included in the final solution.
- VI. Any Enhancements and future work.

B. Codes

C. Presentation:

- I. Business part.
- II. Technical part.

Project Schedule:

Phase	Week	Due Date
Team Formation	Week 7	Friday 27 th March, 11:59 PM
Project Proposal	Week 8	Thursday 2 nd April, 11:59 PM
Final Delivery	Week 13	TBD

Notes:

- There is a penalty for late submissions in any of the three mentioned phases.
- **Any sign of cheating or plagiarism will not be tolerated and will be graded ZERO in the project.**

Suggested Ideas:

1. Kaggle competitions and especially active ones

<https://www.kaggle.com/competitions>

2. Kaggle datasets

<https://www.kaggle.com/datasets>

3. Analytics Vidhya has some datasets and sometimes competitions

<https://datahack.analyticsvidhya.com/contest/all/>

4. Places to find datasets for data science projects:

<https://www.dataquest.io/blog/free-datasets-for-projects/>

5. Data Science for Social Good: It has ideas but no data.

<http://www.dssgfellowship.org/projects/>